

Walsh restricted isometries under sampling with replacement

An extension of the entropic sampling argument

Working manuscript, 30 September 2026

Abstract

Let $N = 2^d$, and sample m rows of the Walsh matrix independently and uniformly, retaining repetitions. We determine, up to universal constants, the smallest m for which the normalized sample has restricted isometry constant at most $1/2$ at sparsity k with probability at least 0.9 : the answer is one for $k = 1$, and $k \log(2k) \log(2eN/k)$ for $2 \leq k \leq N$. The upper bound uses the entropic encoding estimates of Burstein, Iosevich, and Krause. We extend their sampling argument by applying the decoder to indexed row occurrences and randomizing paired assignments to two independent samples. The lower bound follows the Walsh subspace obstruction of Błasiok, Lopatto, Luh, Marcinek, and Rao. Averaging over affine offsets under the fixed-sample law gives an elementary second-moment bound uniform over all dyadic sparsities. A reduction completes the dense and nondyadic regimes. The entropic encoding result is an explicitly identified input from the public literature; the sampling extension and the uniform lower-bound calculation are proved here.

1 Statement and attribution

All logarithms are natural unless a base is displayed. Let $G = \mathbb{F}_2^d$, $d \geq 1$, and $N = 2^d$. Write

$$H(a, b) = (-1)^{a \cdot b}, \quad a, b \in G, \quad H^T H = N I_N.$$

Thus $N^{-1/2}H$ is the normalized Walsh matrix. Let $a_1, \dots, a_m$ be independent uniform elements of G and set

$$\Phi_m = \frac{1}{\sqrt{m}} \begin{pmatrix} H(a_1, :) \\ \vdots \\ H(a_m, :) \end{pmatrix}. \quad (1)$$

Every draw is retained, including repetitions. Denote by $E_{m,N,k}$ the event that

$$\frac{1}{2} \|x\|_2^2 \leq \|\Phi_m x\|_2^2 \leq \frac{3}{2} \|x\|_2^2 \quad \text{for all } x \in \mathbb{R}^N \text{ with } |\text{supp } x| \leq k, \quad (2)$$

and define

$$m_*(N, k) = \min\{m \in \mathbb{N} : m \geq 1, \mathbb{P}(E_{m,N,k}) \geq 0.9\}.$$

This is the convention in Problem FR-10 of the public *Open Problems in Numerical Linear Algebra* catalog [3].

Theorem 1.1. *For every $N = 2^d$, $d \geq 1$, one has $m_*(N, 1) = 1$. There are universal constants $c, C > 0$ such that, simultaneously for $2 \leq k \leq N$,*

$$c k \log(2k) \log \frac{2eN}{k} \leq m_*(N, k) \leq C k \log(2k) \log \frac{2eN}{k}. \quad (3)$$

One may take $c = 1/2000$. The upper bound uses the entropic encoding estimates of [1, Sections 2–4], in the occurrence form stated in Lemma 2.1 below.

Attribution and scope. The essential improvement in the upper logarithmic rate belongs to Burstein–Iosevich–Krause [1]. Their stated sampling theorems concern Bernoulli selection and uniform subsets of distinct rows. Here we supply the passage to independent sampling with replacement, including $m > N$. This is an extension of their proof, not an independent method for obtaining the improved rate.

The lower-bound mechanism—a missed Walsh subspace produces a sparse kernel vector, followed by a second-moment argument—is due to Błasiok–Lopatto–Luh–Marcinek–Rao [2]. The calculation here uses affine offsets and the exact joint law for a fixed number of independent draws to obtain one uniform, nonasymptotic bound. We make no claim that the underlying obstruction is new. The catalog also records Colbrook’s sharper leading-constant analyses at small fixed sparsities and near full sparsity [3]; the order estimate (3) does not supersede those constants.

The sole specialized external proof input is Lemma 2.1. All other estimates needed below are proved in this manuscript. The references are public; no unpublished companion note or private file is a dependency. In particular, the lower bound is independent of [1]. Statements about the upper bound carry exactly the dependence on that paper identified above. This working manuscript does not assert an exhaustive priority determination for the extensions.

2 The entropic input and its occurrence interpretation

For this section let H be any real $N \times N$ matrix with $H^T H = N I_N$ and $|H(a, b)| \leq 1$. Fix $2 \leq k \leq N$ and put

$$\mathcal{K}_k = \{y \in \mathbb{R}^N : \|y\|_2 = 1, \|y\|_1 \leq \sqrt{k}\}, \quad L = \log \frac{2eN}{k}. \quad (4)$$

For $z_a(y) = |(Hy)_a|^2$, orthogonality and the norm constraint give

$$\mathbb{E}_a z_a(y) = 1, \quad 0 \leq z_a(y) \leq k. \quad (5)$$

Every k -sparse unit vector belongs to $\mathcal{K}_k$.

Choose $0 < \eta < 1/16$. Let $a_j = (1 + \eta)^j$, $b_j = a_j^2$, and choose integers $j_- < j_+$ satisfying

$$\frac{\eta}{4(1 + \eta)} < a_{j_-} \leq \frac{\eta}{4}, \quad \frac{\sqrt{k}}{1 - \eta} \leq a_{j_+} < \frac{(1 + \eta)\sqrt{k}}{1 - \eta}.$$

Write $\mathcal{J} = \{j_-, \dots, j_+\}$ and $\ell = |\mathcal{J}|$. Elementary geometric summation gives

$$\ell \leq C_\eta \log(2k), \quad B := \sum_j b_j \leq C_\eta k, \quad (6)$$

$$\sum_{j \leq h} b_j \leq C_\eta b_h, \quad \sum_j b_j (j_+ - j + 1) \leq C_\eta k. \quad (7)$$

Constants C_η may depend only on η ; constants denoted C_0 below are absolute.

Lemma 2.1 (Entropic encoding input; Burstein–Iosevich–Krause). *Fix an ordered list of M rows $h_1, \dots, h_M$ of H , with repetitions allowed. For every $y \in \mathcal{K}_k$, there are correction records with d_j positions and one of three labels at each position at level j , such that*

$$0 \leq d_j \leq M, \quad D := \sum_j b_j d_j \leq C_\eta k L. \quad (8)$$

A deterministic decoder produces disjoint subsets $I_j \subseteq [M]$ and $v_i = \sum_j b_j \mathbf{1}_{I_j}(i)$ satisfying

$$|v_i - |h_i y|^2| \leq C_0 \eta |h_i y|^2 + C_0 \eta^2. \quad (9)$$

The set I_j is determined by the records at levels $h \geq j$. At each level, the number of possible records of length d is at most

$$N_M(d) = \binom{M}{d} 3^d, \quad 0 \leq d \leq M. \quad (10)$$

The decoder can also be run on arbitrary such records, whether or not they arise from a target y .

Justification of the occurrence interpretation. The estimates and decoder are the public input from [1, Lemmas 2.2 and 3.1, Proposition 3.2, and Lemmas 4.1–4.2 and 6.1]. Their deterministic procedure uses an ordered sequence of bounded row queries. Replace each query index by its occurrence index. The experts remain $\{\pm \sqrt{k} e_b : b \in [N]\}$, so the initial comparison entropy remains bounded by L , independent of M . Each recorded correction at level j decreases the potential by at least $3\eta^2 b_j / (32k)$. Telescoping proves (8) for the repeated-query sequence. The decoder follows the same query order and updates, so its amplitude estimate and the higher-level dependence identified in [1, Lemma 6.1] are unchanged. Record positions are now chosen from $[M]$, giving (10). These steps use neither distinctness of queries nor $M \leq N$. $\square$

The distinction between M and N will be important: M counts observed occurrences, whereas N counts coordinates in the target. No enlargement of the ambient coordinate space is involved.

3 From entropic records to two independent samples

The estimates in this section implement the occurrence extension of the counting and symmetrization argument of [1, Sections 5–8]. We give the probability calculation explicitly because signs attached to paired occurrences need not be independent across individual occurrences.

3.1 Weighted counts

Set $h_M(d) = \log N_M(d)$, with $h_M(0) = 0$. If $D = \sum_j b_j d_j > 0$, the binomial estimate and concavity of the logarithm give

$$\begin{aligned} \sum_j b_j h_M(d_j) &\leq \sum_{d_j > 0} b_j d_j \log \frac{3eM}{d_j} \\ &\leq D \log \frac{3eMB}{D}. \end{aligned} \quad (11)$$

Indeed, apply Jensen with weights $b_j d_j / D$; the corresponding weighted average of $1/d_j$ is at most B/D . If $D = 0$, the left side is zero. Since $D \leq MB$ and $u \mapsto u \log(3eMB/u)$ is increasing on $(0, MB]$, (8) and (6) imply, for $M = 2m$,

$$\sum_j b_j h_M(d_j) \leq C_\eta k L \log \left(e + \frac{m}{L} \right). \quad (12)$$

For completeness, let $D_0 = C_\eta k L$ be the bound in (8). If $D_0 \leq MB$, substitute D_0 in (11); the factor MB/D_0 is at most a constant depending on η times m/L . If $D_0 > MB$, use $D \log(3eMB/D) \leq MB \log(3e) \leq D_0 \log(3e)$. This proves (12) in both cases.

3.2 Paired signs and simultaneous decoded bands

Fix an ordered pool

$$\Omega = \{(t, 0), (t, 1) : 1 \leq t \leq m\}$$

of $M = 2m$ row occurrences. A row value may appear more than once. Independently of this fixed pool, let $\varepsilon_1, \dots, \varepsilon_m$ be independent uniform signs. For a deterministic subset $I \subseteq \Omega$, put

$$S_I = \sum_{t=1}^m \varepsilon_t (\mathbf{1}_I(t, 0) - \mathbf{1}_I(t, 1)).$$

Writing the coefficient in parentheses as c_t , one has

$$\sum_t c_t^2 \leq |I|, \quad \mathbb{E}_\varepsilon e^{u S_I} = \prod_t \cosh(u c_t) \leq e^{u^2 |I|/2}. \quad (13)$$

Here $\log \cosh x \leq x^2/2$, as follows by integrating $\tanh x \leq x$ for $x \geq 0$ and using symmetry. Chernoff's argument therefore yields, for $q > 0$,

$$\mathbb{P}_\varepsilon \{|S_I| > \sqrt{2|I|q}\} \leq 2e^{-q}. \quad (14)$$

For an empty set the asserted bound holds deterministically.

Lemma 3.1. *Fix $0 < \rho < 1$. With conditional probability at least $1 - \rho$ over the paired signs, simultaneously for all targets $y \in \mathcal{K}_k$ and the records of Lemma 2.1,*

$$\left| \sum_t \varepsilon_t (v_{t,0} - v_{t,1}) \right| \leq \sqrt{2V A_m}, \quad V = \sum_{i \in \Omega} v_i, \quad (15)$$

where one may take

$$A_m = C_\eta \left[kL \log \left(e + \frac{m}{L} \right) + k \log \frac{2\ell}{\rho} \right]. \quad (16)$$

The constant can be chosen independently of the pool and of m, N, k .

Proof. Let

$$Z_M = \sum_{d=0}^M N_M(d)^{-1} \leq \sum_{d=0}^M 3^{-d} \leq \frac{3}{2}, \quad \pi_M(d) = \frac{1}{Z_M N_M(d)}.$$

For a fixed j and tail profile $\mathbf{d} = (d_j, \dots, d_{j+})$, the decoded sets I_j constitute a family $\mathcal{G}_{j,\mathbf{d}}$ of cardinality at most

$$\prod_{h \geq j} N_M(d_h).$$

This family is fixed before the signs are drawn. Allocate failure probability

$$p_{j,\mathbf{d}} = \frac{\rho}{\ell} \prod_{h \geq j} \pi_M(d_h)$$

to it. Summing these probabilities over all profiles and levels gives ρ . For nonempty families, a union bound using (14) shows that all their bands satisfy $|S_I| \leq \sqrt{2|I|Q_{j,\mathbf{d}}}$, outside an event of probability at most ρ , where

$$Q_{j,\mathbf{d}} = \log \frac{2|\mathcal{G}_{j,\mathbf{d}}|}{p_{j,\mathbf{d}}}$$

$$\leq \log \frac{2\ell}{\rho} + 2 \sum_{h \geq j} h_M(d_h) + (j_+ - j + 1) \log Z_M. \quad (17)$$

Empty families require no bound.

Now take the records of any target y . By disjointness of its decoded bands and Cauchy-Schwarz,

$$\begin{aligned} \left| \sum_t \varepsilon_t(v_{t,0} - v_{t,1}) \right| &\leq \sum_j b_j \sqrt{2|I_j|Q_{j,\mathbf{d}}} \\ &\leq \sqrt{2 \left(\sum_j b_j |I_j| \right) \left(\sum_j b_j Q_{j,\mathbf{d}} \right)}. \end{aligned}$$

The first parenthesis equals V . For the second, reverse the order in the double sum of (17) and apply (6)–(7) to obtain

$$\sum_j b_j Q_{j,\mathbf{d}} \leq C_\eta \left[k \log \frac{2\ell}{\rho} + k + \sum_h b_h h_M(d_h) \right] \leq A_m.$$

The last step is (12), with the constant enlarged. The union bound covered all records, so this event is uniform in y , even if the target is selected after the signs have been observed. $\square$

3.3 The independent-copy inequality

Let $a_{t,0}, a_{t,1}$, $1 \leq t \leq m$, be independent uniform row indices. Apply the preceding lemma to this ordered pool, and use the independent signs to assign one member of each pair to the first sample and the other to the second. The two resulting samples are independent lists of m uniform rows. For $y \in \mathcal{K}_k$, let their energies be X_y and X'_y . Regardless of the assignments,

$$X_y + X'_y = \sum_{t,s} z_{a_{t,s}}(y), \quad X_y - X'_y = \sum_t \varepsilon_t(z_{a_{t,0}}(y) - z_{a_{t,1}}(y)).$$

By (9),

$$V \leq (1 + C_0\eta)(X_y + X'_y) + 2C_0\eta^2 m.$$

The difference between the signed sums of v and z is at most $C_0\eta(X_y + X'_y) + 2C_0\eta^2 m$. Lemma 3.1 therefore proves the following statement after averaging over the pool.

Proposition 3.2. *For two independent samples of m uniform rows, with probability at least $1 - \rho$,*

$$\begin{aligned} |X_y - X'_y| &\leq C_1 \sqrt{A_m(X_y + X'_y + \eta^2 m)} \\ &\quad + C_1\eta(X_y + X'_y) + C_1\eta^2 m \quad (y \in \mathcal{K}_k), \end{aligned} \quad (18)$$

where C_1 is absolute and A_m is as in (16). There is no restriction $m \leq N$.

The decoder's occurrence order must be fixed before the signs. In particular, it is not obtained by first listing whichever occurrences the signs assign to the first sample. Keeping the original paired order is what makes the families in the union bound deterministic.

4 The with-replacement upper bound

Lemma 4.1 (Absorption). *Fix $0 < \delta < 1/4$. There exist positive constants c_0, c_1 , depending only on the absolute constant C_1 in (18), such that the following holds. Suppose $u, v \geq 0$, $m > 0$, $|v - m| \leq \delta m/4$, $\eta \leq c_0 \delta$, $A \leq c_1 \delta^2 m$, and*

$$|u - v| \leq C_1 \sqrt{A(u + v + \eta^2 m)} + C_1 \eta(u + v) + C_1 \eta^2 m.$$

Then $|u - m| \leq \delta m$.

Proof. Put $s = u + v$. Since $s \leq 2v + |u - v|$ and $v \leq (1 + \delta/4)m$, Young's inequality bounds the square-root term by $s/4 + C_2 A + C_2 \eta m$, for an absolute C_2 . Choose η small enough that $C_1 \eta \leq 1/4$. Using $A \leq m$ then gives $s \leq C_3 m$, with C_3 absolute. Substitute this bound in the assumed inequality to get

$$|u - v| \leq C_4 \sqrt{A m} + C_4 \eta m.$$

Choose c_0, c_1 small enough that the right side is at most $3\delta m/4$. The triangle inequality proves the result. $\square$

Theorem 4.2. *For every fixed $0 < \delta < 1/4$, there is $C_\delta > 0$ such that*

$$m \geq C_\delta k \log \frac{2eN}{k} \log(2k) \tag{19}$$

implies, with probability at least $23/24$,

$$\left| \frac{1}{m} \sum_{t=1}^m |(Hy)_{a_t}|^2 - 1 \right| \leq \delta \quad \text{for every } y \in \mathcal{K}_k.$$

This holds for any real H as above and independent uniform row draws.

Proof. Set $\rho = 1/32$ and first fix $\eta > 0$ small enough for Lemma 4.1. We claim that a sufficiently large C_δ in (19) ensures $A_m \leq c_1 \delta^2 m$. Indeed, put $t = m/(kL)$ and $q = \log(2k)$. By (6) and (16),

$$\frac{A_m}{m} \leq C_\eta \left[\frac{\log(e + kt)}{t} + \frac{\log(64\ell)}{Lt} \right]. \tag{20}$$

Here $L \geq \log(2e)$, $\ell \leq C_\eta q$, and $\log(e + kt) \leq q + \log(e + t)$. If $t \geq Bq$, the right side of (20) tends uniformly to zero as $B \rightarrow \infty$: $q/t \leq 1/B$, $\log(e + t)/t$ is decreasing, and $\log(64C_\eta q)/q$ is bounded for $q \geq \log 4$. This proves the claim.

For a fixed $y \in \mathcal{K}_k$, the independent-copy energy satisfies, by (5),

$$\mathbb{E} X'_y = m, \quad \text{Var}(X'_y) \leq mk.$$

Consequently Chebyshev's inequality gives

$$\mathbb{P}\{|X'_y - m| > \delta m/4\} \leq \frac{16k}{\delta^2 m} \leq \frac{1}{4} \tag{21}$$

once $m \geq 64k/\delta^2$, also ensured by increasing C_δ .

Let B be the event that the first sample fails the asserted uniform estimate. On B , choose a witness y with $|X_y - m| > \delta m$. There is no measurability problem: the first sample takes only finitely many values, and a witness can be chosen for each bad value. Conditional on the first sample, (21)

applies to this fixed witness. Thus with conditional probability at least $3/4$, its independent-copy energy lies within $\delta m/4$ of m . On that event, Lemma 4.1 implies that (18) cannot hold for this witness. Proposition 3.2 therefore gives

$$\frac{3}{4}\mathbb{P}(B) \leq \rho, \quad \mathbb{P}(B) \leq \frac{4\rho}{3} = \frac{1}{24}.$$

This proves the theorem, with no upper restriction on the number of draws. $\square$

5 A uniform affine-subspace lower bound

Return to the Walsh matrix on $G = \mathbb{F}_2^d$. Fix

$$K = 2^r, \quad 1 \leq r \leq d-1, \quad s = d-r. \quad (22)$$

For an r -dimensional subspace $V \subseteq G$ and a linear functional $f \in V^*$, define

$$A(V, f) = \{a \in G : a \cdot v = f(v) \text{ for all } v \in V\}.$$

This affine row subspace has density $1/K$.

Lemma 5.1 (A missed affine subspace). *If none of the m sampled rows lies in $A(V, f)$, then Φ_m has a K -sparse unit vector in its kernel.*

Proof. Define $x_b = K^{-1/2}(-1)^{f(b)}$ for $b \in V$ and $x_b = 0$ otherwise. Then $\|x\|_2 = 1$. Orthogonality of characters on V gives

$$(Hx)_a = K^{-1/2} \sum_{b \in V} (-1)^{a \cdot b + f(b)} = \sqrt{K} \mathbf{1}_{A(V, f)}(a).$$

The sampled coordinates of this vector vanish under the stated condition. $\square$

Let X count all pairs (V, f) for which $A(V, f)$ is missed. The number of such pairs is $T = K \begin{bmatrix} d \\ r \end{bmatrix}_2$, where the bracket denotes a Gaussian binomial coefficient. In the fixed- m sampling model,

$$p = (1 - 1/K)^m, \quad \mathbb{E}X = Tp > 0. \quad (23)$$

Lemma 5.2 (Exact offset average). *Fix r -dimensional subspaces V, W with $\dim(V \cap W) = j$. Average over independent uniform $f \in V^*$ and $g \in W^*$. The probability that both $A(V, f)$ and $A(W, g)$ are missed is*

$$2^{-j} \left(1 - \frac{2}{K} + \frac{2^j}{K^2}\right)^m + (1 - 2^{-j}) \left(1 - \frac{2}{K}\right)^m. \quad (24)$$

For $j = 0$, this is exactly p^2 .

Proof. The restrictions of f and g to $V \cap W$ agree with probability 2^{-j} . In that case their common affine solution set has codimension $\dim(V + W) = 2r - j$, hence density $2^j/K^2$; otherwise it is empty. Each of the two affine sets has density $1/K$. A single draw misses their union with the probabilities appearing in (24). Independence of the m draws gives the powers. At $j = 0$ the first base equals $(1 - 1/K)^2$, and the second term has coefficient zero. $\square$

Lemma 5.3 (Intersection probability). *Let V, W be independent uniform r -dimensional subspaces of $\mathbb{F}_2^d$, and write $q_j = \mathbb{P}\{\dim(V \cap W) = j\}$. For $1 \leq j \leq r$,*

$$q_j \leq 4 \cdot 2^{-j(s-r+j)}. \quad (25)$$

Every feasible j also satisfies $j \geq r - s$.

Proof. Fix V . If $\dim(V \cap W) \geq j$, then some j -dimensional subspace $U \subseteq V$ lies in W . For a fixed U , elementary subspace counting gives

$$\mathbb{P}\{U \subseteq W\} = \prod_{i=0}^{j-1} \frac{2^r - 2^i}{2^d - 2^i} \leq 2^{-js}.$$

For example, the product identity follows by counting ordered independent j -tuples in a random W and in the whole space. A union bound over the $\begin{bmatrix} r \\ j \end{bmatrix}_2$ possible U yields

$$q_j \leq \begin{bmatrix} r \\ j \end{bmatrix}_2 2^{-js} \leq 4 \cdot 2^{j(r-j)} 2^{-js}.$$

For the last inequality, use

$$\begin{bmatrix} r \\ j \end{bmatrix}_2 = 2^{j(r-j)} \prod_{i=0}^{j-1} \frac{1 - 2^{i-r}}{1 - 2^{i-j}} \leq \frac{2^{j(r-j)}}{\prod_{h=1}^{\infty} (1 - 2^{-h})} < 4 \cdot 2^{j(r-j)}.$$

Indeed the infinite product is at least $(1 - 1/2)(1 - 1/4)(1 - \sum_{h \geq 3} 2^{-h}) = 9/32 > 1/4$; the inequality for the tail follows first for finite products and then by a limit. Finally $\dim(V + W) \leq d$ implies $j \geq 2r - d = r - s$. $\square$

Proposition 5.4 (Uniform second moment). *Under (22), if m is a positive integer satisfying*

$$m \leq \frac{Krs \log 2}{100}, \quad (26)$$

then

$$\frac{\mathbb{E}X^2}{(\mathbb{E}X)^2} \leq 1 + \frac{4}{2^{3/4} - 1} < 7. \quad (27)$$

In particular, with probability greater than $1/7$, Φ_m has a K -sparse unit vector in its kernel.

Proof. Using Lemma 5.2 to average over all ordered pairs of affine subspaces, divide by p^2 from (23). Put

$$A_j = 1 + \frac{2^j - 1}{(K - 1)^2}, \quad B_0 = 1 - \frac{1}{(K - 1)^2} \in [0, 1].$$

The exact ratio is

$$R := \frac{\mathbb{E}X^2}{(\mathbb{E}X)^2} = \sum_{j=0}^r q_j \left[2^{-j} A_j^m + (1 - 2^{-j}) B_0^m \right]. \quad (28)$$

Since $A_0 = 1$ and $B_0^m \leq 1$,

$$R \leq 1 + \sum_{j \geq 1} q_j 2^{-j} (A_j^m - 1)$$

$$\leq 1 + \sum_{\substack{j \geq 1 \\ q_j > 0}} q_j 2^{-j} \exp\left(\frac{m 2^j}{(K-1)^2}\right). \quad (29)$$

Here and below $j \leq r$ in sums over feasible intersections.

Write $l = r - j$. Feasibility gives $0 \leq l \leq s$ and $j \geq 1$. The elementary inequality

$$rs 2^{-l} \leq 2j(s-l+1) \quad (30)$$

holds throughout this range. For $l = 0$ it is immediate. For $l \geq 1$,

$$\frac{s}{s-l+1} \leq l, \quad \frac{r}{j} = 1 + \frac{l}{j} \leq l+1,$$

so the ratio of the left side of (30) to $j(s-l+1)$ is at most $l(l+1)2^{-l} \leq 2$. The last inequality holds at $l = 1, 2$ and propagates for $l \geq 2$ because $(l+2)/(2l) \leq 1$.

As $K \geq 2$, one has $K/(K-1)^2 \leq 4/K$. Combining this with (26) and (30) yields

$$\begin{aligned} \frac{m 2^j}{(K-1)^2} &\leq \frac{4 \log 2}{100} rs 2^{j-r} \\ &\leq \frac{8 \log 2}{100} j(s-r+j+1) \\ &\leq \frac{\log 2}{4} j(s-r+j+1). \end{aligned} \quad (31)$$

Lemma 5.3 now gives, for every term of (29),

$$q_j 2^{-j} \exp\left(\frac{m 2^j}{(K-1)^2}\right) \leq 4 2^{-\frac{3}{4}j(s-r+j+1)}.$$

Every feasible j has $s-r+j+1 \geq 1$. Summing a geometric series proves

$$R \leq 1 + 4 \sum_{j \geq 1} 2^{-3j/4} = 1 + \frac{4}{2^{3/4} - 1} < 7.$$

The final strict inequality follows from $2^{3/4} > 5/3$, which can be checked by raising both positive sides to the fourth power. Finally, Cauchy–Schwarz applied to $X \mathbf{1}_{\{X > 0\}}$ gives

$$\mathbb{P}\{X > 0\} \geq \frac{(\mathbb{E}X)^2}{\mathbb{E}X^2} > \frac{1}{7}.$$

Lemma 5.1 proves the kernel assertion. □

Remark 5.5. The offset average in (24) is taken under the actual independent-draw law. No Poissonization or independent occupation approximation is used. In particular, $j = 0$ contributes exactly one to the normalized pair probability. For highly overlapping subspaces, (30) controls the excess without a restriction that r or $d - r$ grow with d . This is why a constant failure probability suffices uniformly up to the boundary of the dyadic range.

6 All sparsities and completion of the theorem

Proof of Theorem 1.1. Each column of Φ_m has squared norm exactly one for every $m \geq 1$. Thus $m_*(N, 1) = 1$.

For $2 \leq k \leq N$, Theorem 4.2 with $\delta = 1/8$ gives (2) with probability at least $23/24 > 0.9$ whenever $m \geq Ck \log(2k) \log(2eN/k)$. Taking a ceiling and enlarging the universal constant proves the upper bound in (3).

For the lower bound suppose first that $N \geq 4$. Choose

$$r = \min\{\lfloor \log_2 k \rfloor, d - 1\}, \quad K = 2^r.$$

Then $k/2 \leq K \leq k$ and $2 \leq K \leq N/2$. Proposition 5.4 excludes every positive integer

$$m \leq T_0 := \frac{K \log K \log(N/K)}{100 \log 2}$$

from the defining set for $m_*(N, k)$: a K -sparse kernel vector is also k -sparse, and its probability exceeds 0.1. Consequently $m_*(N, k) > T_0$. If $T_0 < 1$, this conclusion follows instead directly from $m_*(N, k) \geq 1$.

Let $F = k \log(2k) \log(2eN/k)$. Since $k \leq 2K$, $K \geq 2$ and $N/K \geq 2$,

$$\log(2k) \leq 3 \log K, \quad \log \frac{2eN}{k} \leq \log \frac{2eN}{K} \leq 4 \log \frac{N}{K}.$$

Thus $F \leq 24K \log K \log(N/K)$ and

$$T_0 \geq \frac{F}{2400 \log 2} > \frac{F}{2000}.$$

For the remaining case $N = 2$, $k = 2$, the inequality $m_*(2, 2) \geq 1$ is more than sufficient for the same lower constant. This completes the proof for every parameter pair. $\square$

Remark 6.1 (Full sparsity and multiplicities). At $k = N$, the formula gives order $N \log N$. In fact, if n_a denotes the number of times row a is sampled, orthogonality diagonalizes $\Phi_m^T \Phi_m$ with eigenvalues $N n_a / m$. Hence the full-sparsity event requires every occupancy to lie between $m/(2N)$ and $3m/(2N)$. Keeping all N distinct rows would give an exact isometry, but it is a different sampling rule. The proof above retains the prescribed rule also when $m > N$.

Remark 6.2 (What the extension establishes). The result is an order estimate at fixed distortion and fixed success probability. It does not determine the optimal upper constant, sharp distortion dependence, or the leading constants in endpoint asymptotics. Its upper-bound mechanism is the entropic encoding of [1]. The additional work is the probability interface for repeated occurrences and the uniform affine second-moment completion of the Walsh lower bound. Those are the contributions claimed here.

References

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